

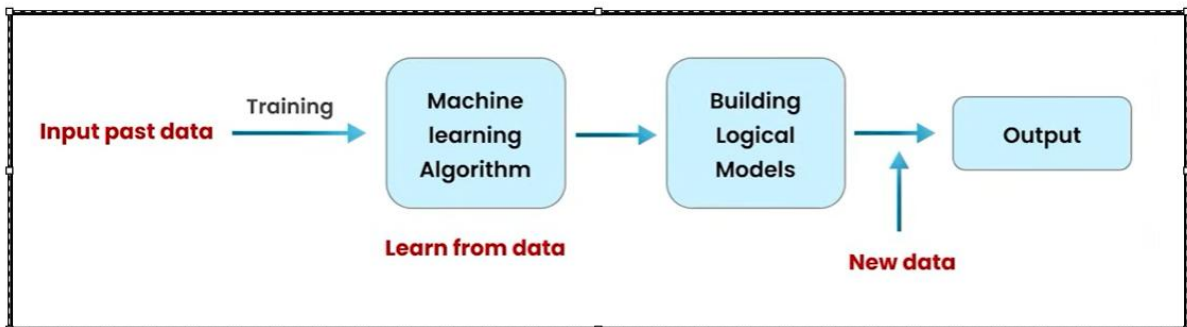
CHAPTER 1 : Computational foundation of Machine Learning

1.1 Introduction of Machine Learning :

1.1.1 Definition :- Machine Learning is a field of study that gives computers the ability to learn from data and improve performance without being explicitly programmed.

1.1.2 Features of Machine Learning :

- Data-Driven approach
- Model based learning
- Learning from experience
- Pattern recognition
- Mathematical optimization



1.1.3 Need of Machine Learning :

Machine Learning is needed because traditional programming is not sufficient to handle complex, large scale and dynamic problems.

- **Handling large amount of data**
 - huge amount of data is generated social media , banking , sensors .
 - It is impossible to analyze this data manually.
 - Machine Learning helps in automatic data analysis.
- **Solving complex problem**
 - Some problems cannot be solve using fixed rules.
 - Machine Learning learns patterns instead of using hard-coded logic.

- **Improves accuracy over a time**
 - Machine Learning learn from new data
 - Performance improves automatically with experience
- **Automation decision making**
 - Machine learning enables automatic predictions and decision.
 - Example : Fraud detection, Recommendation system.
- **Cost and time efficiency**
 - Reduce human efforts.

1.1.4 History of Machine Learning :

1936 – Alan Turing formalized the theory of computation.
 1950 – Alan Turing proposed the Turing Test for machine intelligence.
 1957 – Frank Rosenblatt developed the Perceptron learning algorithm.
 1986 – Backpropagation enabled training of multi-layer neural networks.
 2006 – Revival of Deep Learning approaches.
 2012 – Major breakthrough in large-scale image recognition using Deep Learning.
 2017 – Introduction of Transformer architecture for modern AI systems.

1.2 Machine Learning VS Traditional programming :

1.2.1 Traditional Programming :

- Programmer write explicit rules.
- Logic is manually defined.
- Does not improve automatically with new data.
- Input data + Program -> Output.
- Example : Even or Odd number

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If number % 2 == 0    -> Even
Else                  -> Odd
```

1.2.2 Machine Learning :

- System learn from data.
- Logic is learned automatically.
- Improve performance with more data.

- Input data + output (data) -> Program (model)

- Example :

Data :

X1	X2	Y
2	1	2
2	2	4
2	3	6
2	4	?

Model learn pattern :

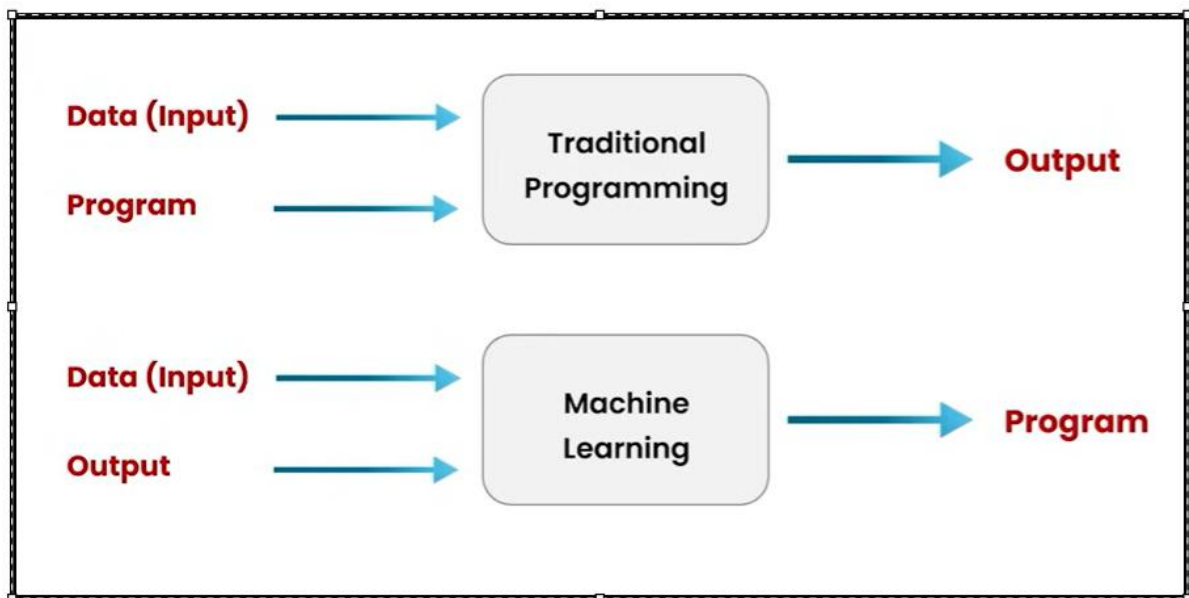
$$Y = X1 * X2$$

New Input :

$$X1 = 3, X2 = 5$$

Predicted Output :

$$Y = 15.$$



1.3 Relationship between Artificial Intelligence , Machine Learning , Deep Learning :

1.3.1 Artificial Intelligence (AI) :

Artificial Intelligence is the branch of computer science that enables machine to perform tasks that normally require human intelligence .

Such as , learning , reasoning , decision making , problem solving .

Artificial Intelligence includes :

- Rule-based system
- Knowledge representation and reasoning
- Optimization techniques
- Logical inference mechanism

1.3.2 Machine Learning (ML) :

Machine Learning is a field of study that gives computers the ability to learn from data and improve performance without being explicitly programmed.

Machine Learning based on :

- Statistical modeling
- Probability theory
- Loss function minimization
- Model generalization

1.3.2 Deep Learning :

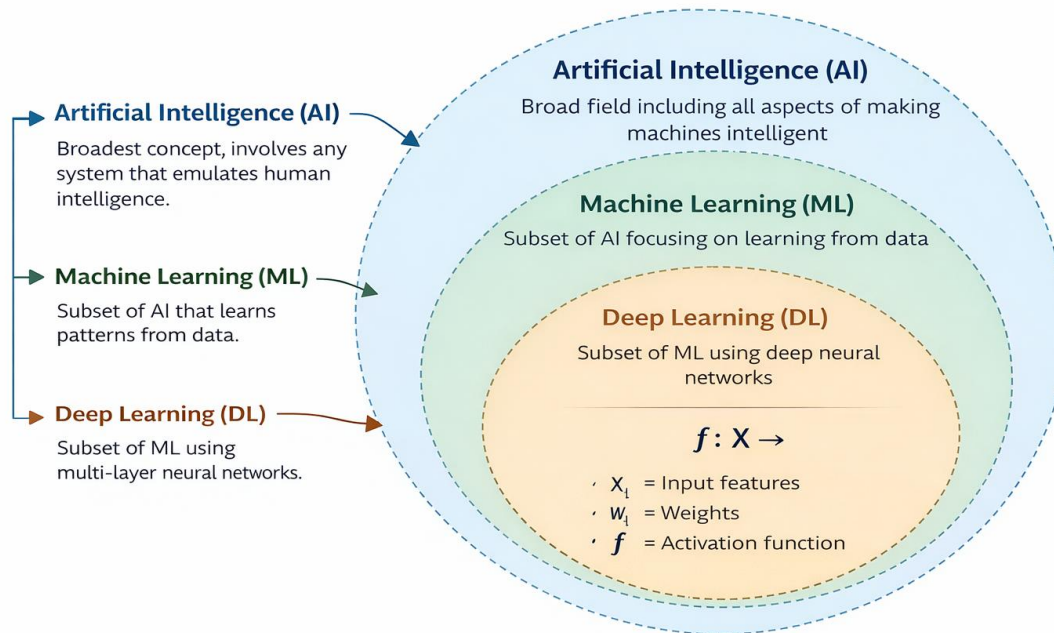
Deep learning is a subset of Machine learning that utilizes “ multi-layer neural networks ” to learn complex problem from large scale data.

A neural networks computes :

- x_i = Input features
- w_i = Weights
- b = Bias
- f = Activation function

Deep Learning automatically extracts hierarchical feature representation for raw data and is widely use in computer vision , speech recognition and natural language processing.

Relationship between Artificial Intelligence, Machine Learning, and Deep Learning



1.4 Types of problems solved using Machine Learning :

Machine Learning algorithm is designed to solve different categories of problems based on the nature of data and learning objective.

These problems are generally classified into predictive , descriptive and design-making tasks.

1.4.1 Classification Problem :

Classification is a supervised learning problem in which the output is discrete or categorical value .

Output type :

- Categorical(Yes/NO , 0/1 , Spam/ not Spam).

Examples :

- Email spam detection(Spam / not Spam)
- Disease Diagnosis (Positive / Negative)

1.4.2 Regression Problems :

Regression is supervised learning problem in which the output is a continuous numeric value.

Output type :

- Continuous value (Price, Temperature, Salary)

Examples :

- House price prediction
- Stock price forecasting
- Temperature prediction

1.4.3 Clustering Problems :

Clustering is an unsupervised learning problem used to group similar data points together without labeled output.

Output type :

- Groups or clusters

Examples :

- Customer segmentation
- Market basket analysis
- Document grouping

1.4.4 Dimensionality Reduction :

Dimensionality reduction is used to reduce the number of input features while preserving important information.

Examples :

- Feature selection in high-dimensional datasets
- Data compression
- Principal Component Analysis (PCA)

1.4.5 Anomaly Detection :

Anomaly detection identifies rare or unusual data points in a dataset.

Examples :

- Fraud detection
- Network intrusion detection
- Manufacturing defect detection

1.4.6 Recommendation Problems :

Recommendation systems predict user preferences based on past behavior.

Examples :

- Product recommendation
- Movie recommendation
- Content recommendation

1.5 Learning Systems and Problem Formulation :

1.5.1 Learning Systems :

Learning System is a system that learns from data and improves its performance automatically.

1.5.1 A Machine Learning system consists of :

1.5.1.1 Data :

- Collection of input and output.
- X = Input features
- Y = Target output

Example :

- X = Area, Rooms
- Y = House Price

1.5.1.2 Model :

- A mathematical function that maps input to output.
- $f: X \rightarrow Y$

Example :

- $\text{Price} = w_1(\text{Area}) + w_2(\text{Rooms}) + b$
- Model contains parameters (weights and bias).

1.5.1.3 Loss Function :

- Loss function measures the difference between actual output and predicted output.
- $\text{Loss} = (y - \hat{y})^2$

1.5.1.4 Optimization :

Optimization algorithm updates model parameters to reduce loss.

1.5.2 Problem Formulation :

Problem Formulation means defining a real-world problem in mathematical form so that a Machine Learning model can solve it.

Problem formulation includes :

- Define Input (X)
- Define Output (Y)
- Identify type of problem
- Define objective function
- Select evaluation metric

Example :

Problem : Predict student pass or fail.
Study hours, Attendance

Output (Y) : Pass / Fail
Objective : Minimize classification error
Evaluation : Accuracy

1.6 Real-World Constraints in Machine Learning :

In real-world applications, Machine Learning systems face several practical limitations and constraints.

1.6.1 Data Quality Issues :

Machine Learning depends heavily on data.

Real-world data may be :

- Incomplete
- Missing values
- Imbalanced
- Biased

1.6.2 Limited Data Availability :

Sometimes sufficient labeled data is not available.

Small dataset may cause :

- Overfitting
- Poor generalization

1.6.3 Computational Constraints :

Training ML and Deep Learning models requires :

- High processing power
- Large memory
- GPUs or cloud resources

1.6.4 Time Constraints :

Real-world systems require fast prediction.

Example :

- Fraud detection must respond in milliseconds.
- Self-driving cars need real-time decisions.

1.6.5 Model Interpretability :

Some complex models are difficult to interpret.

This creates challenges in:

- Healthcare
- Finance
- Legal systems

1.6.6 Deployment and Maintenance :

- Data distribution may change (Data Drift)
- Model performance may degrade
- Continuous monitoring is required

1.7 Scope, Applications, and Limitations of Machine Learning :

1.7.1 Scope of Machine Learning :

Scope refers to the growth potential, research opportunities, and areas where Machine Learning can be applied in the present and future.

Machine Learning is expanding rapidly due to :

- Big Data growth
- High computational power
- Availability of cloud computing

- Development of Deep Learning

Scope Areas :

1.7.1.1 Artificial Intelligence Systems :

Machine Learning forms the foundation of modern Artificial Intelligence systems. While traditional AI relied on rule-based logic, modern AI systems are data-driven.

Example :

- Self-driving cars
- Intelligent assistants
- Robotics

1.7.1.2 Data Science and Analytics :

With the exponential growth of digital data, traditional analytical methods are insufficient. Machine Learning provides scalable techniques to analyze large, high-dimensional datasets.

Applications include:

- Predictive analytics
- Customer behaviour modelling
- Business intelligence systems

1.7.1.3 Automation and Smart Systems :

Machine Learning plays a significant role in industrial automation and smart systems.

Examples :

- Predictive maintenance in manufacturing
- Smart city management

1.7.1.4 Research and Innovation :

Machine Learning continues to evolve through research in areas such as:

- Explainable Artificial Intelligence (XAI)
- Reinforcement Learning
- Transfer Learning
- Federated Learning
- Ethical AI

1.7.2 Applications of Machine Learning :

Applications refer to the practical implementation of Machine Learning techniques in real-world systems.

1.7.2.1 Healthcare :

Machine Learning plays a crucial role in improving medical diagnosis, treatment planning, and drug discovery.

Applications:

- Disease prediction
- Medical image analysis
- Early diagnosis of chronic diseases
- Personalized medicine

Impact :

- Reduces diagnostic errors
- Enables early disease detection
- Supports doctors in decision-making

1.7.2.2 Finance and Banking :

Machine Learning is widely used in financial institutions for risk analysis and fraud prevention.

Applications :

- Fraud detection
- Credit scoring
- Risk assessment
- Stock price prediction
- Algorithmic trading

Example : credit card fraud detection

In credit card fraud detection, ML models analyze transaction features such as:

- Transaction amount
- Location
- Time
- Device used

Impact:

- Reduces financial loss

- Improves security
- Automates risk management

1.7.2.3 Transportation and Autonomous Systems :

Machine Learning enables intelligent transportation systems.

Applications :

- Self-driving cars
- Traffic prediction
- Route optimization

Example : Autonomous vehicles

Autonomous vehicles use deep learning models to :

- Detect lanes
- Identify pedestrians
- Recognize traffic signs

Impact:

- Reduces accidents
- Improves traffic efficiency
- Enhances transportation safety

1.7.2.4 :Natural Language Processing :

Machine Learning enables machines to understand and process human language.

Applications :

- Chatbots and virtual assistants
- Machine translation
- Sentiment analysis
- Speech recognition
- Text summarization

Example : Sentiment analysis models classify customer reviews as positive, negative, or neutral.

Impact :

- Automates customer service
- Improves communication
- Enables intelligent conversational systems

1.7.2.5 Cybersecurity :

Machine Learning helps in detecting cyber threats and anomalies.

Applications :

- Intrusion detection
- Malware classification
- Phishing detection
- Network anomaly detection

Example : An ML model analyzes network traffic patterns. If abnormal behavior is detected compared to normal baseline activity, it triggers an alert.

Impact:

- Enhances system security
- Reduces cyber attacks
- Provides real-time monitoring

1.7.3 Limitations of Machine Learning :

Machine Learning (ML) has transformed many industries, it has several limitations. These limitations must be understood before deploying ML systems in real-world environments.

1.7.3.1 Dependency on Large and High-Quality Data :

Machine Learning models rely heavily on data. The quality and quantity of data directly affect model performance.

Explanation :

If the training data is :

- Incomplete
- Noisy
- Biased
- Incorrect

The model will produce inaccurate predictions.

1.7.3.2 Overfitting and Underfitting :

Machine Learning models may not generalize well.

Overfitting:

Model memorizes training data but performs poorly on unseen data.

Occurs when :

- Model is too complex
- Training data is limited

Underfitting:

Model is too simple and fails to capture underlying patterns.

1.7.3.3 Data Privacy and Security Concerns :

Machine Learning systems often use sensitive data such as:

- Personal information
- Financial records
- Medical history

Improper handling can lead to:

- Privacy violations
- Data breaches
- Ethical concerns

1.7.3.4 Ethical and Bias Issues :

ML systems can unintentionally:

- Reinforce social bias
- Discriminate against certain groups
- Make unfair decisions